RIA ARORA

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Summary

AI Research Engineer interested in **spatial awareness** and **uncertainty estimation** for robotics, with hands-on experience deploying and testing on real robot platforms (manipulators, ground vehicles, and autonomous navigation stacks). Published in *Robotics and Automation Letters* (2025). Have successfully graduated and defended my master's thesis on **uncertainty estimation on non-Euclidean manifolds**. Based in Montreal and authorised to work in Canada.

Skills

Robotics	SLAM, Sensor Fusion, Differentiable Filtering, Localisation, ROS/ROS2, Sensor Calibration, Motion Planning, Robot Manipulation, Navigation Stacks
ML / AI	Uncertainty Estimation, Probabilistic Modeling, Generative Models (GANs, Diffusion, Flows), Density Estimation, Time Series Forecasting, Computer Vision, Optimisation on Non-Euclidean Manifolds
Programming	Python, PyTorch, TensorFlow, JAX, CUDA, C++, Java, RxJava, Scala
Deployment	Spark, Hadoop, Azure, Kubernetes, Springboot, Hive, Jenkins, Qt toolkit
Data Storage	Redis, Elasticsearch, AWS Databricks, Couchbase, Kafka

Experience

Probing FinGPT's Forecasting Module | Blog| Code — Montreal, Canada
Independent Research Project — *2026*

- Benchmarked FinGPT's unpublished **forecasting module** against classical (ARIMA, XGBoost) and neural (GPT-4, base LLaMA) baselines on **Dow 30** over a 2-week horizon; fine-tuned LLaMA-3 led all baselines on directional accuracy (61.2%) while its MSE remained near a driftless random walk, isolating genuine directional signal from magnitude forecasting.
- Designed **masking ablations** (numerical values vs. financial vocabulary, with random-substitution controls) to test whether the model reasons over numeric content or qualitative sentiment language, finding that fine-tuning induces **joint representations** across both modalities.
- Fine-tuned LLaMA-2 on **Chain-of-Thought traces** to test whether explicit reasoning improves forecasts; found marginal directional gains at the cost of magnitude accuracy, with generated traces often narrating output format rather than reasoning over financial content.

Quebec AI Institute (MILA) — Montreal, Canada
Research Assistant — *September 2023 – December 2025*

- Developed a novel **differentiable Bayesian filter** for robot pose estimation that models non-Gaussian sensor noise using non-parametric probability distributions learned end-to-end from data.
- Achieved more **calibrated posterior estimates** over robot pose compared to Gaussian baselines; framework generalises to object detection, depth estimation, and other perception tasks.
- GitHub | Thesis | Webpage

SLAM in Degenerate Feature Environments — Montreal, Canada
Independent Research Project — *2024 – Present*

- Investigating **sensor adaptation for SLAM** in degenerate feature environments (featureless corridors, low-texture outdoor terrain) using a **LiDAR-Inertial-Visual** sensor suite (LiDAR, IMU, RGB-D).
- Building on **LIO-SAM** and **Super Odometry** backbones; extending to **visual-inertial odometry** for navigational robots (ROS2 Humble) to improve robustness when LiDAR features degrade.
- Modelling and propagating **localisation uncertainty** under uninformative observations, bridging directly to thesis work on non-Gaussian uncertainty estimation on Lie groups.

Robotics and Embodied AI Lab (REAL) — Montreal, Canada
Research Assistant — *April 2023 – August 2023*

- Contributed to the **Harmonic Exponential Filter (HEF)** for non-parametric Bayesian state estimation on Lie groups, using Fourier transforms for efficient multimodal distribution convolution.
- Work accepted to the IROS workshop and subsequently published in **Robotics and Automation Letters (RAL)**, January 2025.

- Hands-on experience with the **ViperX 150** robotic arm using **ROS1** and **MoveIt**: trained and deployed pick-and-place policies, implemented learned inverse kinematics, and integrated the full motion planning pipeline.
- Conducted hands-on experiments with real robot platforms including **Clearpath Husky** and **AgileX Bunker** ground vehicles, including sensor calibration and ROS2 integration.
- Worked on **Duckietown** autonomous vehicle platforms, building an intersection navigation system using inter-robot communication, kinematics, projective geometry, motion planning, and control.

Bosch Engineering Centre, Assisted Automotive Driving — Cluj-Napoca, Romania
Software Development Engineer — *September 2021 – September 2022*

Smart Data Team – Computer Vision — *November 2021 – September 2022*

- Built an **active learning pipeline** to surface high-value unlabelled examples for computer vision model training, reducing annotation cost.
- Developed similarity-based retrieval models using **Siamese triplet loss** and FAISS with EfficientNet embeddings – a key component in RAG-style information retrieval pipelines.
- POC on object tracking across sequences (occlusion handling, redundancy reduction) and on GAN-based image augmentation using semantic label maps (OASIS) to bootstrap first-iteration models.
- Introduced dependency management and code packaging practices to the team.

Metadata Management for Endurance Runs — *September 2021 – May 2022*

- Built ETL pipelines in Scala over Hadoop for endurance-run metadata; migrated to Azure Databricks, improving throughput and reducing infrastructure overhead.
- Designed and owned an API overlay service to forecast storage requirements from pipeline metadata, from product requirement through system design.

TESCO UK E-Commerce — Bangalore, India
Graduate Software Development Engineer — *August 2019 – August 2021*

Customer Order — *March 2020 – August 2021*

- Developed transactional REST APIs in a reactive framework for the full order lifecycle on the Tesco online platform, serving **~100k active orders per day**.
- Built custom horizontal pod autoscaling in Kubernetes to eliminate event-listener lag in the asynchronous order journey.
- Implemented a webhook for seamless customer management at order level; improved API robustness for graceful failure handling.

Search Science / Platform — *August 2019 – February 2020*

- Improved relevancy and ranking of product catalogue search queries; contributed to fuzzy (secondary) search improvements worth approximately **£4–5M per year**.
- Containerised infrastructure to enable independent testing environments.

Romanian Institute of Science and Technology (RIST) — Cluj-Napoca, Romania
Undergraduate Research Assistant — *January 2019 – July 2019*

- Implemented Vanilla GAN, Wasserstein GAN, and FGAN from scratch in TensorFlow; benchmarked using FID and Inception Score; explored training stabilisation via hyperparameter tuning.

Solid State Physics Laboratory, DRDO (Min. of Defence) — Delhi, India
Intern — *May 2017 – July 2017*

- Developed Smith-chart software for input/output matching topologies in amplifier design, implementing an auto path-finder that accelerated the research and fabrication workflow.

Education

Université de Montréal & Mila, Quebec AI Institute — Montreal, Canada
M.Sc. (Research) in Artificial Intelligence — *September 2022 – February 2026*

- CGPA: 3.7/4.0   Advisors: Liam Paull and Guy Wolf
- Thesis: Differentiable filter for non-Gaussian, non-parametric uncertainty estimation; applications in robot pose estimation, object detection, and depth estimation.
- Coursework projects: Riemannian diffusion models on mixed-curvature manifolds; multi-task offline multi-agent RL with latent representations; Lyapunov-based nonlinear control; Duckietown intersection navigation with inter-robot communication.

- Academic Experience: ROS/ROS2 integration, sensor calibration, technical writing, literature review, and research presentation.

Birla Institute of Technology and Science, Pilani (BITS) Goa, India
Bachelor of Engineering (Hons.) in Computer Science *August 2015 – May 2019*

- CGPA: 8.27/10 Awarded Excellence Scholarship for academic performance.
- Teaching Assistant for Computer Programming in C++ and Computer Architecture.

Achievements and Interests

- Third place in a Tesco hackathon for improving fuzzy matching in online search Nov 2019
- Captain of the college mixed Ultimate Frisbee team; led team to nationals Aug 2017 – May 2018
- Ranked 2127 out of 100,000 in the IIT JEE Advanced entrance exam 2015
- Enthusiastic about music (ukulele), podcasts/audiobooks, and sports (table tennis, badminton, tennis, cycling, ultimate frisbee).
- Avid reader; favourite books include *Crime and Punishment*, *Educated*, *The History of Bees*, *Pride and Prejudice*, *A Man Called Ove*, and *Tuesdays with Morrie*.